

Operation Threshold

Debt-Burden Risk & First-Payment Default Diagnostic

DIVISION	TOOL	DATA PERIOD	PORTFOLIO SIZE
Credit Risk	Excel — Power Query / Power Pivot / DAX	Jan – Dec 2025	~20,000 active loans

BACKGROUND

Vantage Trust Bank operates 10 branches, offering Personal, SME, Auto, and Mortgage loans to retail and small-business customers through a network of 40 loan officers. The credit team approves loans primarily on credit score and income verification, but has never systematically checked whether loan size relative to income predicts repayment failure.

THE INCIDENT

The Head of Credit Risk pulled a routine sample of defaulted accounts for a write-off review and noticed something informally: a disproportionate number of the defaults looked "over-borrowed" relative to what the customer earned. No one has confirmed this with actual data. The Risk Committee wants to know, before the next lending cycle's approval criteria are set, whether debt burden actually predicts default — and whether high-burden borrowers fail immediately or gradually.

"We're approving loans on credit score and income verification, but nobody has checked whether size-of-loan-versus-income is quietly doing more damage than either. I don't want a guess. I want a number."

YOUR ROLE

You have just joined Vantage Trust Bank as a Junior Data Analyst on the Credit Risk team. You've been handed raw, unclean loan and repayment extracts covering the full 2025 portfolio. Using Excel — Power Query, Power Pivot, and DAX — you must clean and model the data, calculate each loan's debt burden, and determine whether high-burden loans default more, and whether they fail immediately at first payment or gradually over time.

FLAGGED SIGNALS (UNVERIFIED)

Debt-to-income skew Some loans may be sized far beyond what the customer's income supports	Immediate vs. gradual failure Defaults may cluster at first payment (underwriting failure) or later (repayment stress)
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Product concentration

Certain loan types may carry disproportionately high debt-burden ratios

Branch / officer inconsistency

Some branches or officers may be approving higher-burden loans than others

ANALYTICAL DELIVERABLES

#	DELIVERABLE	LEVEL
1	Portfolio Overview & Baseline KPIs Total loans, total value, overall default rate — establish the baseline before diagnosing.	BEGINNER
2	Data Cleaning & Model Build Clean raw extracts in Power Query; build a fact/dimension star schema in Power Pivot.	BEGINNER
3	Debt-Burden Band Analysis Band Debt-to-Income ratio into Low/Moderate/High/Severe; compare default rate per band.	BEGINNER / INTERMEDIATE
4	First-Payment Default Diagnostic Flag whether defaults occur at first payment vs. later; cross-tab against DTI band.	INTERMEDIATE
5	Branch, Officer & Product Risk Comparison Identify which branches, officers, or products carry the highest-burden, highest-default loans.	INTERMEDIATE
6	Recommendation Dashboard & Written Recommendation Combined DTI-band + FPD dashboard with slicers; authored recommendation on where to tighten approval criteria.	INTERMEDIATE